

Quality / CAPA & Complaints

A Governed Agentic AI Reference Architecture for Life Sciences

Agent 05 of 8 · draft root-cause and CAPA, score complaints — a quality owner approves classification and closure

HCLS AI Agent Suite

HCLS AI Agent Suite · Built on AWS · June 2026

FROM REPEAT FINDINGS TO ROOT CAUSE

Quality / CAPA & Complaints

A governed quality copilot that triages complaints, drafts root-cause analysis and CAPA records, and flags MDR reportability — while a quality owner approves every classification, reportability call, and closure.

#1 cited

CAPA (21 CFR 820.100) is consistently the most-cited FDA device 483 clause

[gov/peer-reviewed — FDA]

561

drug-program 483s issued in FY2024 (211.192 cites up ~171% YoY)

[gov/peer-reviewed + sector-press]

\$10M–\$100M

typical pharma recall cost depending on scope

[industry-research]

The Issue & The Cost of Doing Nothing

THE ISSUE

- Product-complaint volumes are large; CAPA root-cause consistency is a recurring inspection finding.
- On-time CAPA closure rates are a perennial 483/warning-letter theme.
- Complaint handling (820.198) is among the most error-prone QSR areas for device makers.
- FDA's QMSR (effective Feb 2026) raises the bar on quality-record integrity and traceability.

THE COST OF DOING NOTHING

\$10M–\$100M / recall

Industry: a pharma recall runs \$10M–\$100M by scope; non-routine device quality events are estimated at \$2.5B–\$5B/yr industry-wide. CAPA and complaint handling are the perennial top inspection findings.

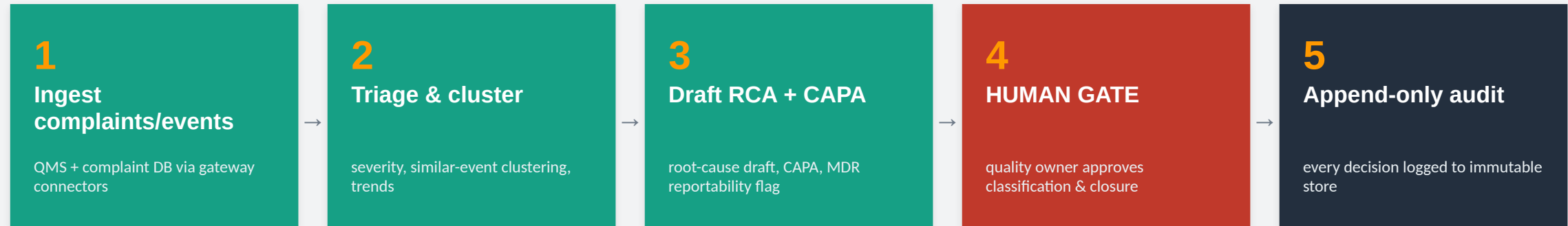
- Warning letters and consent decrees from repeat root-cause and closure findings.
- Recall and MDR exposure from late or mis-classified complaints.

[industry-research — recall-cost analysis; FDA inspection data]

The design line: the agent drafts root-cause, CAPA, and reportability recommendations; a quality owner approves classification, MDR reportability, and closure.

How We Solve It — A Governed Pipeline

Read complaints and quality events through a scoped gateway, draft root-cause and CAPA and flag reportability — then route every disposition to a quality owner.



Every action audited

node, timestamp, actor, data sources, model + prompt version — append-only and inspection-ready (21 CFR Part 11).

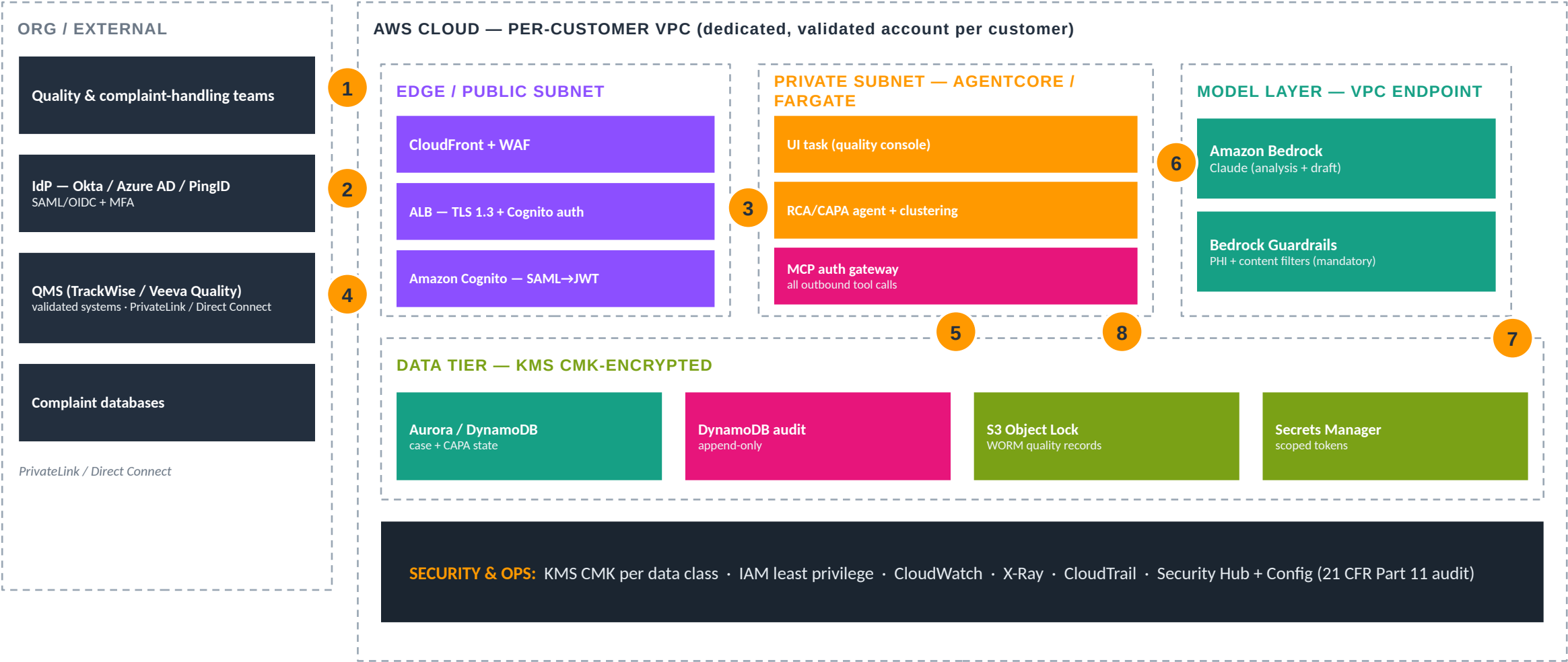
Grounded & explainable

every regulated claim traces to the source corpus; Bedrock Guardrails mask PHI before any model call and bound the output.

AI never decides the CAPA disposition or reportability

the consequential action is a human-only step, enforced in the orchestration framework — not a policy SOP.

AWS Architecture & Traffic Flow



1 user sign-in 2 SAML federation (MFA at IdP; Cognito issues JWT) 3 authenticated session to quality console 4 QMS + complaint-DB reads over private connectivity 5 demand scales the agent workers 6 model calls stay in-VPC via Bedrock + Guardrails 7 every classification, RCA, and closure persisted to append-only audit 8 connectors reachable only through the governed MCP gateway

Proof, Payback & How to Deploy

MEASURED OUTCOMES

consistency

structured RCA vs. variable manual root-cause

[design property]

#1 finding

CAPA is the clause this directly addresses

[gov/peer-reviewed — FDA]

reportability

MDR flag drafted; quality owner decides

[design property]

QMSR-ready

audit + traceability posture for Feb 2026

[gov/peer-reviewed]

WHAT IT TAKES TO DEPLOY

1. Provision KMS CMK + per-customer validated VPC / network.
1. Stand up Cognito SAML→JWT federation (analyst / quality-owner roles).
1. Front with CloudFront + WAF, ALB TLS 1.3.
1. Deploy the agent + grant tools via the MCP gateway (deny-by-default).
1. Connect TrackWise/Veeva Quality + complaint DB.
1. Enable S3 Object Lock + append-only audit; run smoke + HITL test.

Deploy: CloudFormation quick-deploy provisions the isolated environment + dual MCP gateway; point connectors at the QMS and complaint database.

aws-native-reference/05-quality-capa/DEPLOY.md · docs/DEPLOY-QUICKSTART.md

The takeaway: the agent isn't the product — the governed platform that makes it 21 CFR Part 11 / GxP-defensible and deployable on AWS is.